

OCTriage: which OCT segmentations should graders read first?

Two Flywheel gears that tell graders what to read first: an independent case study for the Wisconsin Reading Center by Kaan Mutlu

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Why it matters

At WRC, graders review what AI does ([A-EYE update, 2025](#)), although most masks need little or no editing; OCTriage tells them which to read first, steering their effort toward likely errors. In WRC's GA study, graders reviewed every AI mask and found 84.8% needed no or minor edits ([Bogost et al. 2025](#)). WRC's OCT layer model reaches Dice 0.90–0.95 on trial B-scans, and graders review and edit its output as needed ([Balasubramanian et al., SIIM 2026](#)).

Order matters because readers tire: after a day of reading, radiologists found fractures less accurately, their eyes focused less well, and they felt more tired ([Krupinski et al. 2010](#)). In 2022 WRC showed that an AI score and a cutoff can send likely image-label errors to people: on their test set, 27.6% of images would be reviewed and errors fell from 11.7% to 1.5%, and they estimated grader time at 25% of a full manual pass ([Domalpally, Slater et al. 2022](#)). OCTriage applies that idea to OCT segmentation, but every scan is still read and only the order changes: it ranks each visit, then the B-scans inside it.

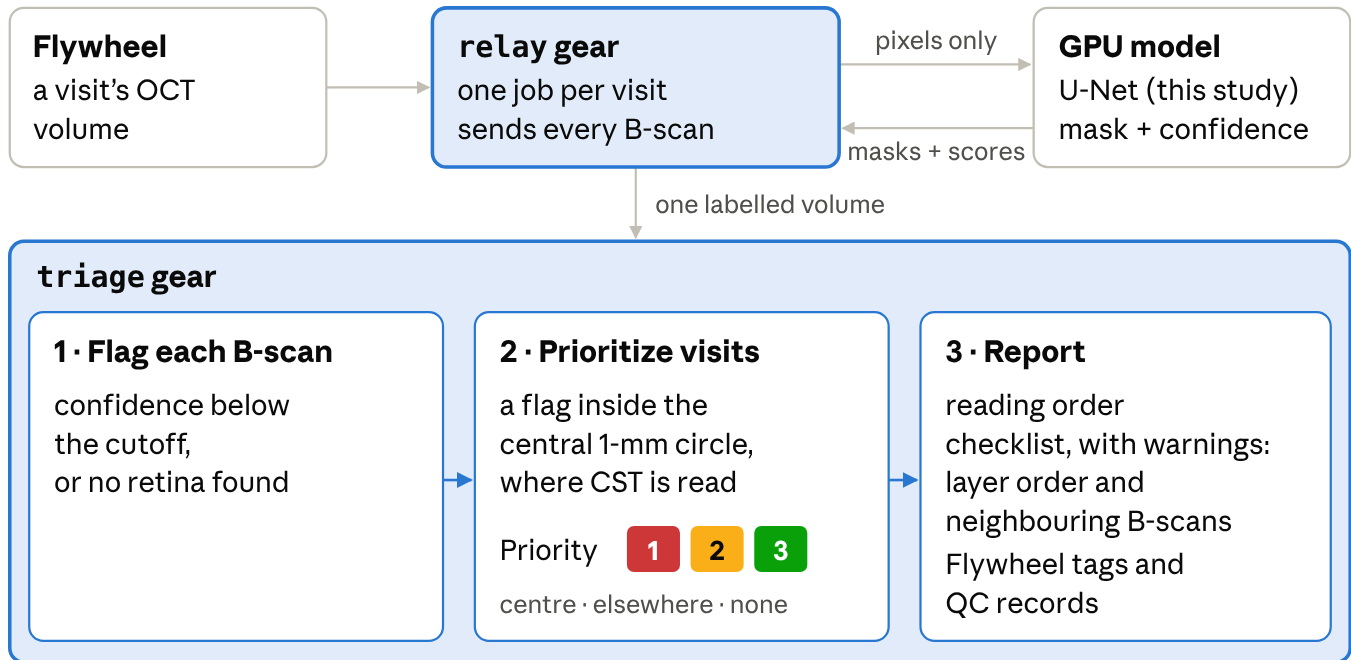
Checking the machine is not new at reading centres: in 2009 WRC found that 30% of 6,741 trial OCTs needed their machine-measured centre-point thickness remeasured by hand ([Domalpally et al. 2009](#)). What is new is deep learning at scale. WRC's models "are not meant to replace human graders, but to support them" ([A-EYE update, 2025](#)), and the FDA's 2025 draft guidance on AI in drug development asks that such a system be judged as a human–AI team, not as a model alone ([FDA 2025](#)).

Getting a model to the graders is its own problem: review boards limit where patient data may go, and clinicians won't use a tool that starts at a command prompt ([Slater 2024](#)). At WRC the data live on Flywheel. Slater's proof of concept showed that a Flywheel gear can call a GPU server's API ([Slater 2026](#)), and WRC's GPUs sit inside UW, on [Platform R](#). OCTriage's relay gear is built to carry the images along that route.

What I built

OCTriage is two Flywheel gears, Docker containers that Flywheel runs on its data, and a GPU model service (Figure 1).

One job per visit: images go to the GPU, a reading order comes back



☐ The two Flywheel gears I built; both ran under Flywheel's own gear runner against the GPU service.

OCTriage · two Flywheel gears and a GPU service

Figure 1. OCTriage's architecture. The two gears (blue) hold no model: **relay** carries images to the GPU and back, and **trriage** turns the answers into a reading order.

1. relay : from Flywheel to a GPU. WRC's Flywheel runs its gears without GPUs ([Slater 2026](#)). Flywheel can add GPU compute on request ([Flywheel docs](#)), but WRC already owns GPUs on Platform R, and a service keeps the model loaded. So **relay** sends the images to a GPU service: the next step of Slater's [fw_model_serving](#) proof of concept, whose idea is his. One job per visit sends every B-scan and stores the masks, each image's confidence and its measurements in one labelled volume. The service keeps the model in GPU memory between requests, so each B-scan costs about 0.17 s rather than a container start and a model load.

Only pixels leave Flywheel: the request carries no names, IDs or dates, though any text burned into an image would travel with it. In the design they go to a GPU inside UW, on [Platform R](#), which keeps patient information to the minimum necessary ([HHS](#)).

2. triage : what to read first. It ranks at two levels. Each visit gets a reading priority (Figure 2): Priority 1 when a flagged B-scan crosses the central 1-mm circle where central subfield thickness (CST) is measured, or when the scan is off-centre; Priority 2 when flags lie elsewhere; Priority 3 when there are none. Inside a visit, a B-scan is flagged when its confidence falls below a cutoff or its mask shows no retina, and flagged B-scans come first, least confident first.

CST steers trials: in a major DME drug trial (DRCR Protocol T), it helped decide who could join and when an eye got another injection ([Wells et al. 2015](#)). Layer-order and neighbouring-B-scan checks appear on the checklist as warnings.

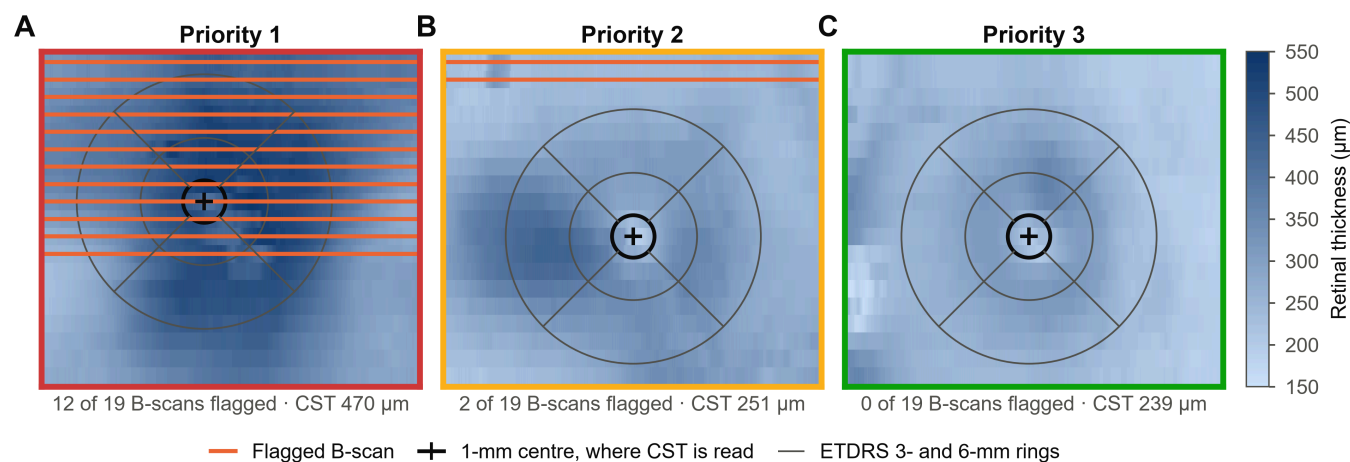


Figure 2. The priority rule on three real DME visits. Retinal thickness maps seen from above, with the ETDRS grid centred on the fovea, the centre of vision (+); CST is the mean thickness inside its central 1-mm circle ([Pak et al. 2013](#)). (A) Priority 1, patient DME-11: flagged B-scans (orange) cross the central circle. (B) Priority 2, patient DME-14: its two flags lie at the edge of the scan, as in 7 of the 9 Priority 2 visits. (C) Priority 3, patient DME-30: nothing is flagged.

triage writes a checklist per image that says where to look, plus Flywheel tags and quality-control (QC) records, Flywheel's built-in fields for check results. Checks that need data this study lacks, image-quality grades and a second visit, still return a record: "na", with the reason.

Confidence is how much the model agrees with itself (Figure 3). The service segments six altered copies of each B-scan (test-time augmentation; [Wang et al. 2019](#)), maps the masks back, and scores their agreement against the consensus as mean Dice. The cutoff, 0.9876, is Youden's index ([Youden 1950](#)): the best balance between poor masks caught and acceptable masks flagged. In testing it was chosen on four fifths of the patients and applied to the other fifth; at WRC it would be set from a stated reading budget or miss rate.

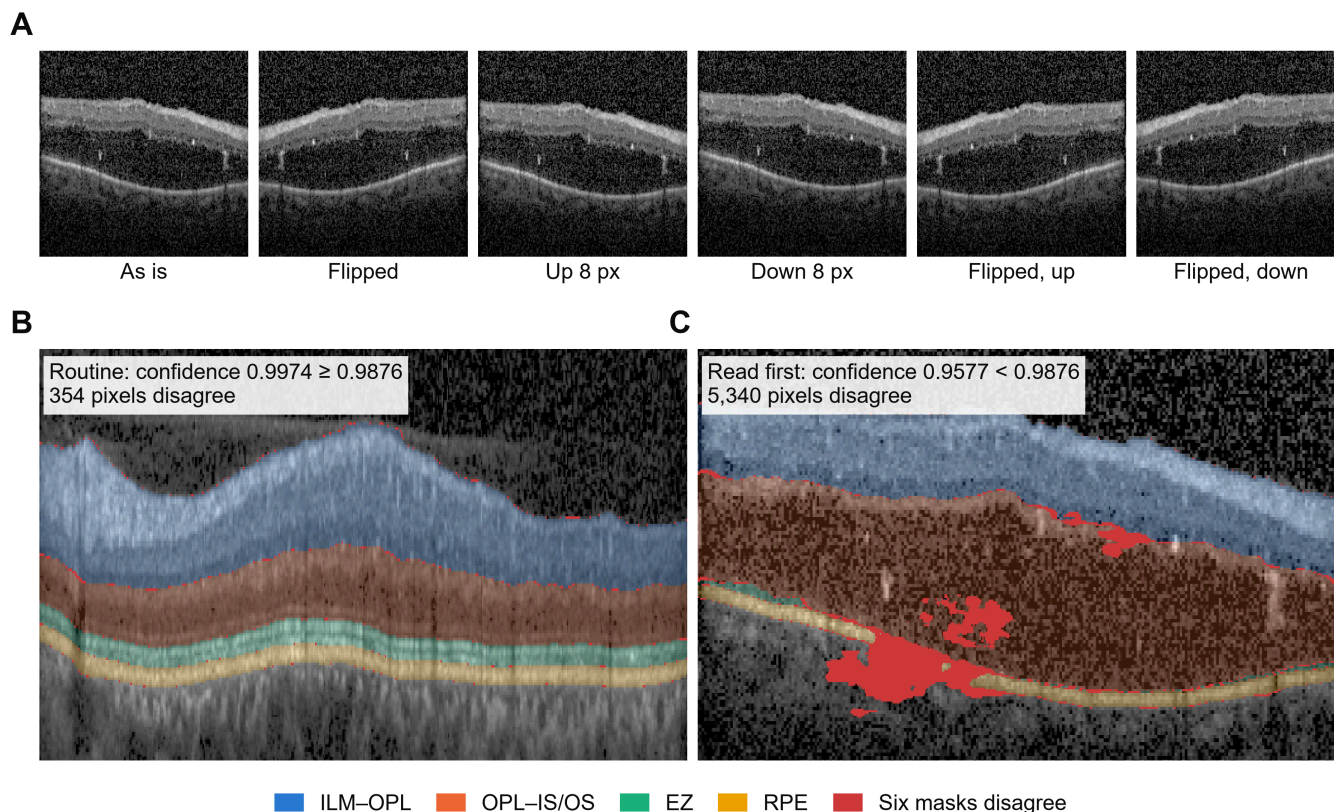


Figure 3. How confidence is measured. (A) The model segments six altered copies of the B-scan. (B) On a routine B-scan the six masks agree almost everywhere. (C) On a B-scan flagged to read first they disagree (red), and its confidence falls below the cutoff.

I trained a 2D U-Net for this study. The gears don't depend on it: any model that returns a mask, a confidence and its measurements fits behind `relay`, WRC's nnSAM included.

Flywheel's own command-line tool runs gear containers on a laptop with the same container contract as a Flywheel site, and needs no site or account. Both gears ran that way on four visits and gave the analysis's results, and automated tests check every rule.

Data: OCT5k, three graders per B-scan

OCTriage was tested on [OCT5k](#), a public OCT dataset (CC0): 1,672 B-scans from 60 patients with age-related macular degeneration (AMD), diabetic macular edema (DME) or healthy eyes, each B-scan traced by three graders. OCT5k stores each B-scan as its own image, so I stacked each patient's B-scans in order into one volume, spaced by the scan size the paper reports. Each patient has one visit: a Spectralis volume of 19 to 61 B-scans over 8.9×7.4 mm (Figure 4).

Cohort	Patients	B-scans	Grader vs grader (Dice)	Model vs graders (Dice)	Poor masks
AMD	21	739	0.916	0.919	21
DME	19	403	0.892	0.892	24
Normal	20	530	0.926	0.932	9
All	60	1,672	0.914	0.917	54

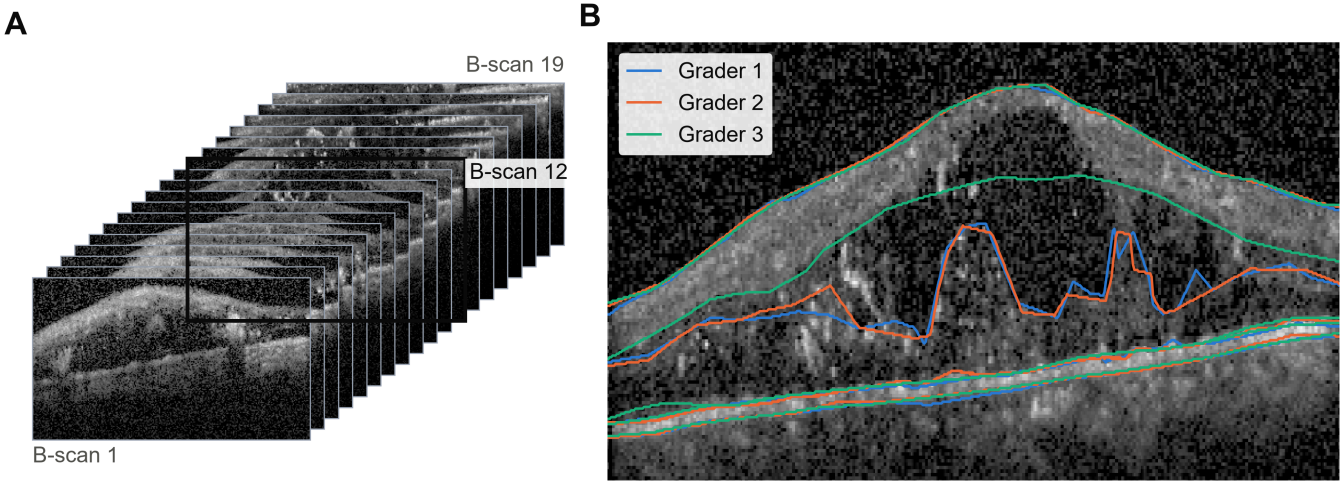


Figure 4. Scans from patient DME-28. (A) The visit is a volume of 19 B-scans; B-scan 12 is outlined. (B) On B-scan 12 the three graders agree on the top of the retina and on the RPE, but not inside the swollen retina.

Every B-scan was scored by a model that never saw its patient: five-fold cross-validation split by patient, as in [Domalpally et al. 2024](#). To score the flags we need an answer key: a mask is poor if it agrees with the graders at least 0.05 Dice less than they agree with each other.

The size of that margin is mine, but the graders back it. A failure threshold is hard to set when inter-grader variability is unknown ([Zenk et al. 2024](#)), and OCT5k measures it: judged the same way, a grader falls more than 0.044 Dice below the other two in only 5% of gradings. So a poor mask is roughly one worse than 95% of human gradings.

What it shows

Read visit by visit in priority order, the Priority 1 visits, 39% of the B-scans, hold all 4 CST errors and 45 of the 54 poor masks; a random order would find about 21 (Figure 5). Inside every visit, the flags mark 8.8% of B-scans and hold 91% of the poor masks.

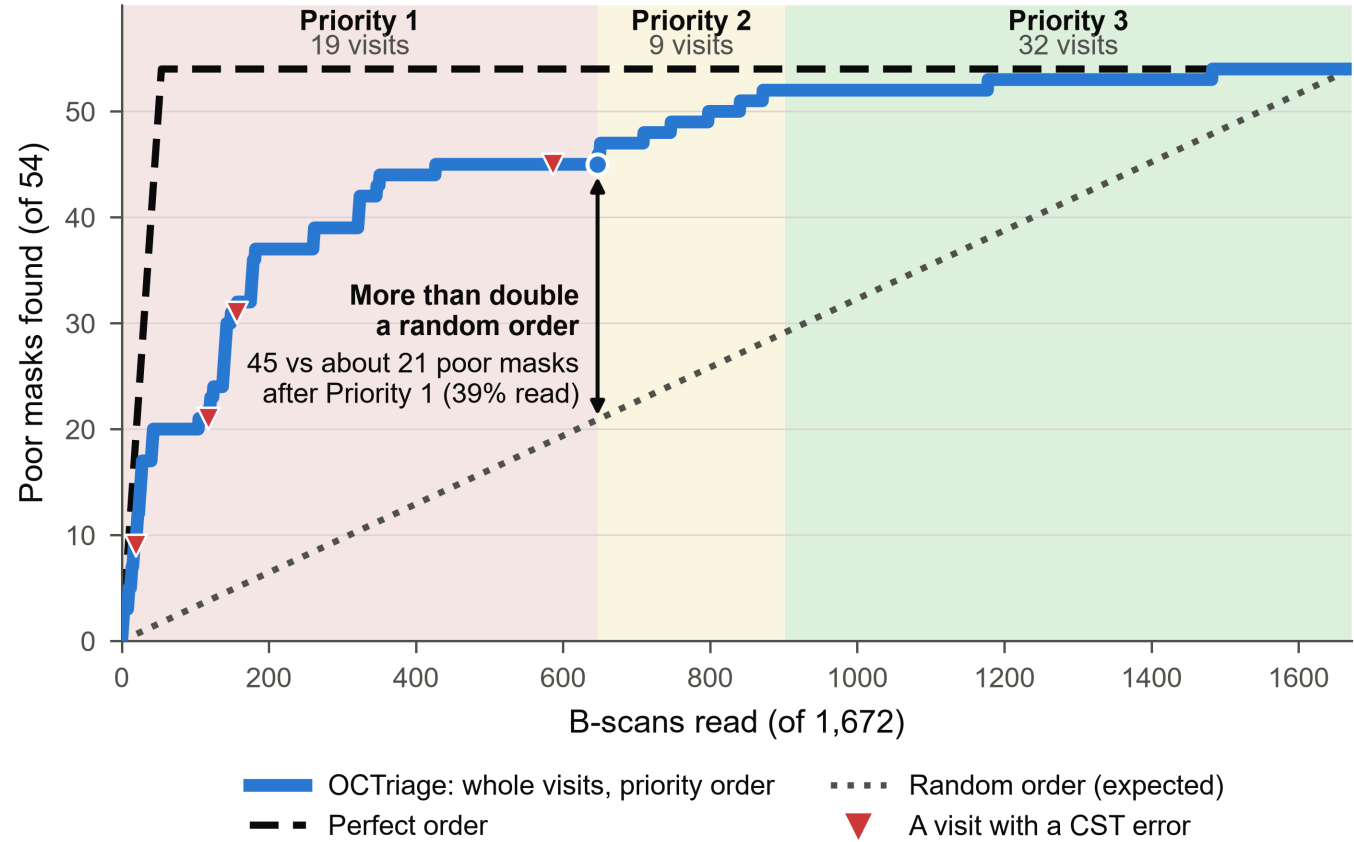


Figure 5. Poor masks found as a grader reads whole visits in priority order, each visit in triage’s order (blue). Shading marks the priorities; triangles mark the four visits where the model’s CST was more than 20.1 μm off the graders’, CST’s repeatability ([Domalpally et al. 2010](#)). Dotted: random order, as expected on average; dashed: perfect order. The arrow marks the gain over a random order once Priority 1 is read.

Measure	Result
B-scans flagged to read first	8.8%
Poor masks among the flagged B-scans	49 of 54: 91%
Poor masks left among the rest	5 of 1,525: 0.33%
Read by the end of Priority 1	39% of B-scans: 45 of 54 poor masks, all 4 CST errors

The result does not hinge on the 0.05 margin: with the graders’ own margin the flags hold 56 of 66 poor masks (85%), and with the strict end of WRC’s $\pm 20\text{--}30\text{ }\mu\text{m}$ reproducibility thresholds ([Balasubramanian et al. 2026](#)) 44 of 55 (80%).

Each comparison's rule was fixed before the results were computed (one declared amendment is in the repository). Two anatomy checks that ignore the model's confidence, layer order and agreement with neighbouring B-scans, caught no more poor masks than the same number of extra confidence-ranked reads, so `triage` shows them as warnings, not as reasons to read first.

The misses matter most. A poor mask in the routine queue is read last, perhaps by a tired grader, under a "routine" label, and readers follow a machine even when it is wrong: very experienced radiologists fell from 82% to 46% correct when an AI suggested the wrong answer ([Dratsch et al. 2023](#)). In a trial a missed error goes into the endpoint, and endpoints decide approvals: in 2025 the FDA approved Encelto on an OCT layer endpoint, the rate of ellipsoid zone (EZ) loss ([FDA 2025](#)).

Conclusion

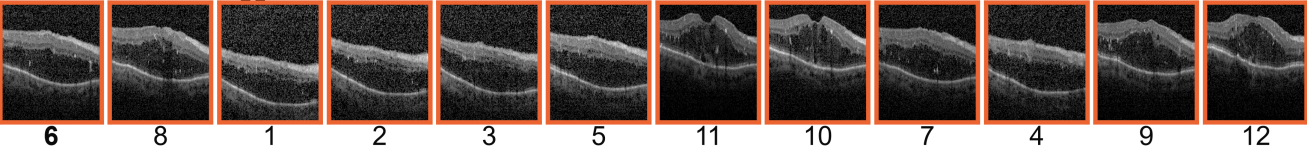
WRC's graders review what AI does ([A-EYE update, 2025](#)), and for trial endpoints that is likely to stay: an OCT measurement was the primary endpoint behind a 2025 drug approval ([FDA, Encelto 2025](#)), and FDA guidance on AI in drug development is still a draft ([FDA 2025](#)). OCTriage keeps full reading and changes only the order: the first 39% of scans held 83% of the poor masks, more than double a random order, so graders meet likely errors first. `relay` works with any model behind the same interface; `triage`'s OCT rules sit in one module, so other imaging would add its own (Future work, item 5). The code, every decision's source and the tests are at github.com/kmutlu21/octriage.

What a grader would see in Flywheel

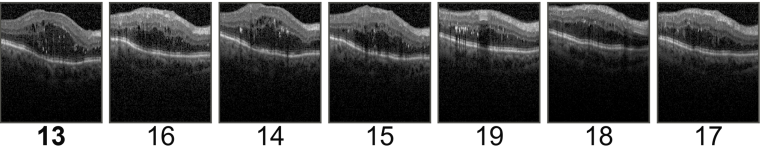
A Priority	Patient	Flagged B-scans	CST (μm)
■ Priority 1 — endpoint at risk	DME-07	19 of 19	440
■ Priority 1 — endpoint at risk	DME-06	18 of 19	230
■ Priority 1 — endpoint at risk	AMD-26	14 of 61	258
■ Priority 1 — endpoint at risk	DME-28	13 of 19	238
■ Priority 1 — endpoint at risk	DME-11	12 of 19	470
	⋮		
■ Priority 2 — flags outside the centre	AMD-21	4 of 61	213
■ Priority 2 — flags outside the centre	DME-14	2 of 19	251
	⋮		
■ Priority 3 — no flags	AMD-08	0 of 19	219

B Patient DME-11 · Priority 1: flagged B-scans 8-10 cross the central 1-mm circle, where CST is measured

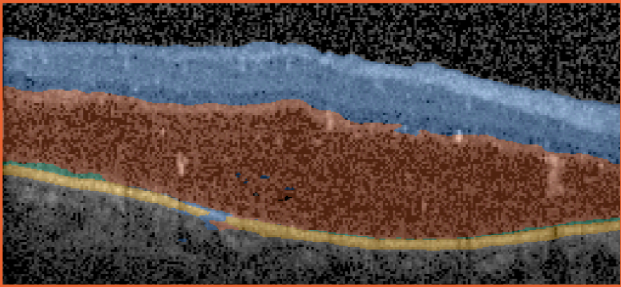
Read first: the 12 flagged, least confident first



Then the other 7

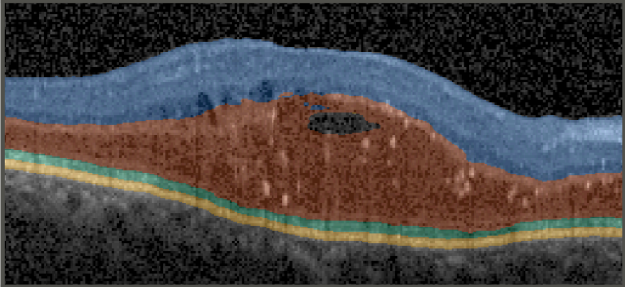


C B-scan 6: read first, 1st in order



- ✗ confidence 0.958 < cutoff 0.988
- ✓ retina found
- ✗ layer order broken (warning),
-1.3 to +1.9 mm from the fovea
- ✓ consistent with its neighbouring B-scans
- EZ not seen in 41%: loss or error?

B-scan 13: routine, 13th in order



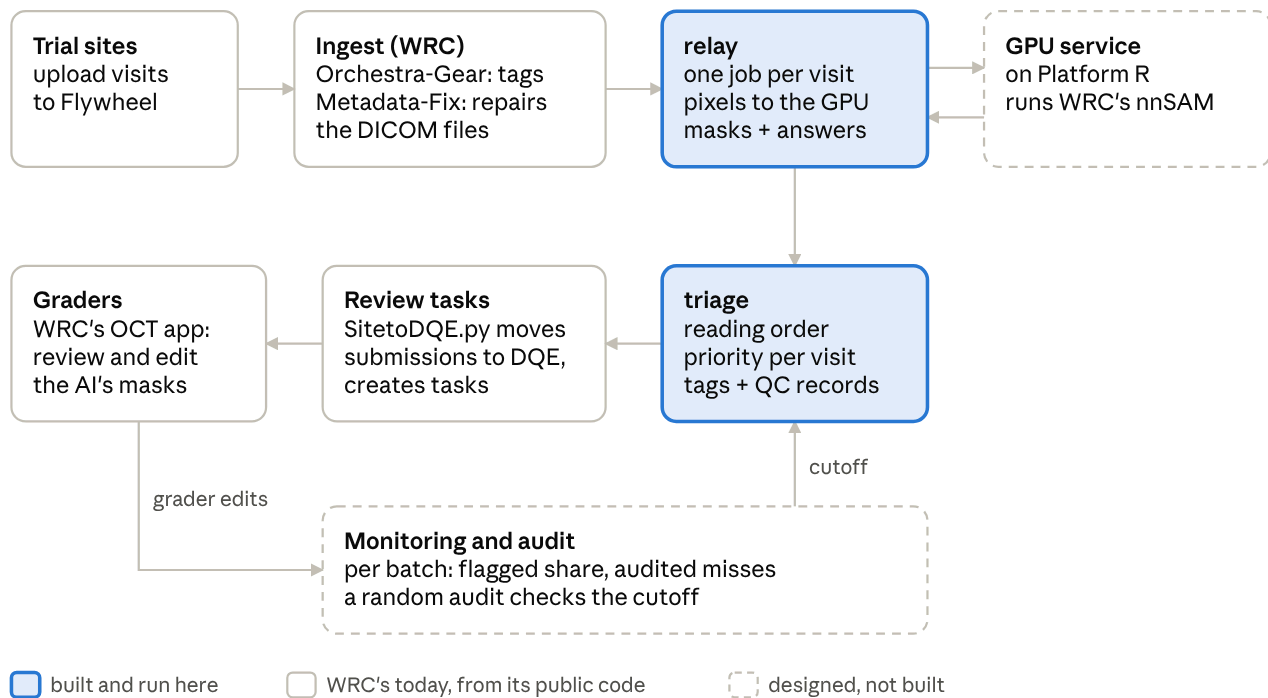
- ✓ confidence 0.992 ≥ cutoff 0.988
- ✓ retina found
- ✗ layer order broken (warning),
+0.5 to +1.6 mm from the fovea
- ✓ consistent with its neighbouring B-scans

Figure 6. What a grader would see in Flywheel: a schematic drawn from the gears' real outputs, not a screenshot. (A) The worklist, Priority 1 visits first. (B) Patient DME-11's 19 B-scans in `triage`'s order, the 12 flagged ones first. (C) A read-first B-scan beside a routine one, each with the model's layers and `triage`'s checklist: B-scan 6, read first, and B-scan 13, read 13th. On trial data the gear would also check image quality and change since the last visit (longitudinal consistency; [Bogost et al. 2025](#)); OCT5k has no quality grades and one visit per patient, so neither check could be tested here.

How it would fit at WRC

Ordering needs the model's answer before a grader opens a case, so both gears would run at ingest (Figure 7). They would follow WRC's [Orchestra-Gear](#), "a gear that adds the correct tags to initiate the automation process in Flywheel", and [Flywheel-Metadate-Fix](#), which repairs ophthalmic DICOM files, and come before [SitetoDQE.py](#), a script that moves site submissions into a review project (DQE) and creates tasks there; all three are by Mohammed Younis. Where OCTriage sits is my interpretation of this public code, not confirmed with WRC; two links I could not check are whether Flywheel can reach Platform R and whether task creation can read a priority.

OCTriage runs at ingest, so the reading order exists before a grader opens a case



OCTriage in WRC's pipeline · built, WRC's today, designed

Figure 7. Where OCTriage would sit in WRC's pipeline. Blue: built and run here; solid grey: WRC's today, from its public code and the SIIM 2026 poster; dashed: designed, not built.

Future work at WRC, and how

1. **Their model, their cutoff.** Put nnSAM behind the same service contract, with confidence from the agreement of its five fold models; an ensemble's agreement was the strongest simple method in [Zenk et al. 2024](#) (there, five networks from different seeds). Learn the cutoff on WRC's graded data, with graders' edit scores as the outcome, such as the four-level score in [Safai et al. 2024](#), and set it from a stated reading budget or miss rate.
2. **A Flywheel test project.** Run `relay` and `trriage` there against a GPU service on Platform R, with encryption and a login on the service (the demo has neither), and check that task creation can use the priority.
3. **The checks this data cannot test.** Image quality first, from WRC's gradability grades; then consistency between visits, as in [Bogost et al. 2025](#). Both return "na" today.
4. **The human-AI team.** Measure graders working with OCTriage, not the model alone ([FDA 2025](#)); show the order without a "routine" verdict, to limit automation bias ([Dratsch et al. 2023](#)); and audit the routine queue: zero misses in about 300 audited B-scans bounds the miss rate below 1% with 95% confidence ([Hanley & Lippman-Hand 1983](#)). OCT5k allows a first rehearsal: each of its three graders in turn corrects B-scans in OCTriage's order, the other two serve as the reference, and the endpoint errors left are counted at each reading budget.
5. **Other models.** `relay` and the core of `trriage` are not tied to OCT. WRC's fundus curation model ([Domalpally, Slater et al. 2022](#)) and its geographic atrophy models on autofluorescence ([Safai et al. 2024](#); [Domalpally et al. 2024](#)) would each need their own check module; that is untested.

Whether a routine queue could one day be sampled rather than read in full is for each trial's charter: the FDA's nonbinding 2018 guidance lets the charter define computer assistance, and suggests that blinded readers verify a subset ([FDA 2018](#)).

Limitations

- **One public dataset.** 60 patients, one visit each, one device, and not trial data. Its graders corrected AI-drawn masks, so their agreement is tighter than it would be for independent tracings.

- **Few endpoint events.** The priority result rests on 4 CST errors and 20 visits with a poor mask.
- **One model.** The U-Net under-reads EZ loss in DME, and any new model needs its own cutoff.
- **No live Flywheel site.** The gears run under Flywheel's own tool with the same container contract; gear rules, tags and the network path on a real site are untested.
- **Sparse volumes.** Most OCT5k volumes have B-scans 308–411 μm apart, so decentration could be judged in only 10 of 60 visits.

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